Quantum Machine Learning for Dark-Matter Substructure

A capacity-matched study on DeepLense strong-lensing images

GSoC 2026 — ML4SCI / DeepLense (QML)

June 14, 2026

The problem & the target

Task. Classify strong-lensing images by dark-matter scenario: axion (vortex) / **cdm** (subhalos) / **no_sub** (smooth).

Classical SOTA. MAE-pretrained ViT — AUC 0.968 (arXiv:2512.06642). The encoder is fine-tuned end-to-end; the head is a single `Linear(192→3)`.

Question. Does attaching / embedding a quantum component beat the *same architecture without the circuit*?

The one rule

Every quantum model is paired with a **capacity-matched classical "sham"** (circuit → classical layer, matched dimensionality). **quantum** − **sham** is the only honest test.

What I tried — 11 distinct directions

#	Direction	Quantum placement
1	Frozen-feature fusion heads (gated/x-attn/QCT/QVF)	readout
2	Pretrain→finetune with quantum head (NAE)	readout
3	QCT / QVF from scratch (end-to-end)	token / encoder
4	Pathology-fixed training (wd=0, U(0,π) init, qlr sweep)	encoder
5	Quantum kernels (fidelity + projected), 8/12/16 qubits	kernel
6	Generative IQP Born machine	generative
7	Quantum anomaly autoencoder (leakage-free)	generative
8	Robustness battery (noise / blur / brightness)	encoder
9	Multi-view quantum fusion (M=1..8)	fusion
10	Quantum inside the ViT encoder (QViT)	mid-encoder
11	Equivariant quantum (C4) 2×2 control	encoder

Results — classical / quantum / sham (discriminative)

Method	Data	Classical	Quantum	Sham
Frozen QCT-head	frozen feats	0.974†	0.9838	0.9838
Frozen X-attn	frozen feats	0.974†	0.9820	0.9822
QFair qct (fixed)	Model_I	0.9633	0.9790	0.9802
QFair qvf (fixed)	Model_I	0.9633	0.9838	0.9830
QCT-scratch	Model_I	0.9633	0.9509	0.9521
QCT-scratch	Model_II	0.9682	0.9756	0.9700
QVF-scratch	Model_I	0.9633	0.9805	0.9790
QVF-scratch	Model_II	0.9682	0.9983	0.9928
QVF-scratch	Dataset1	0.9672	0.9983	0.9960
Dual-enc FiLM qct	Model_I	0.9633	0.9793	0.9806
Dual-enc FiLM qvf	Model_II	0.9682	0.9989	0.9991
QViT (in encoder)	Model_I	0.9670‡	0.9622	0.9697
REQAE equiv (full)	Model_I	0.9788*	0.9768	0.9768**

† shipped linear-probe baseline ‡ plain-ViT (none) $*$ classical-equiv $**$ classical-plain. green=quantum>sham (all single-seed, $\leq +0.006$, at the saturation ceiling).

Results — classical / quantum / sham (other tasks)

Task	Metric	Classical	Quantum	Sham
Multi-view fusion M=8	AUC (Model_I)	0.9796^{c}	0.9788	0.9788
Multi-view fusion M=8	AUC (Model_II)	0.9983^{c}	0.9971	0.9985
Pretrain→ft NAE head	AUC (Model_I)	0.9626	lose0.5028	0.4957
Anomaly detection	AUC (SSL feats)	0.859^{m}	lose0.438	0.571
Generative (NLL↓)	12-bit latent	8.237^{l}	lose8.449	—
Few-shot N=250 equiv	AUC (Model_I)	0.904^{e}	lose0.840	—
Quantum kernel	few-shot AUC	RBF win	loseloses all N	—

c concat fusion *m* Mahalanobis (0-param) *l* matched Ising *e* classical-equivariant. lose = quantum clearly worse than classical/sham.

Across every task: quantum $\leq$ classical/sham, except tiny ceiling-level positives that have not been seed-verified.

Parameter efficiency — sham always has more params

The capacity-matched sham (dimension-matched) ends up with more parameters than the circuit in *every* method:

Method	Quantum	Sham	sham excess
QFair qct	94,555	94,635	+80
QFair qvf	143,091	147,107	+4,016
QCT-scratch	106,571	107,547	+976
QVF-scratch	142,795	144,755	+1,960
QFusion (M=8 fusion core)	1,320	1,704	+384
QViT	210,411	210,795	+384

Honest reading

Where quantum ties (or slightly beats) the sham, it does so with **fewer parameters** $\Rightarrow$ *parameter efficiency at parity* — the legitimate QViT-literature claim, true *without* beating accuracy. Caveat: the one *truly* param-matched control (QAE anomaly, 76 vs 72) had quantum *lose* (0.438 vs 0.496).

Discriminative & training — every arm ties the sham

Method	Quantum	Sham	Δ
Frozen heads (gated/x-attn/QCT)	0.982–0.984	0.982–0.984	≈ 0 tie
QFair qct (pathology-fixed)	0.9790	0.9802	-0.0012 tie
QFair qvf (pathology-fixed)	0.9838	0.9830	$+0.0008$ tie
QCT-scratch (I/II/D1)	.951/.976/.975	.952/.970/.964	$\pm$noise tie
QVF-scratch (I/II/D1)	.981/.998/.998	.979/.993/.996	$+$small tie
QViT (quantum in encoder)	0.962	0.970	-0.008 lose

Key honesty point. An early "$+0.0072$" (QCT, single seed) did not replicate — re-run gave -0.0012. Isolated positives flip sign across seeds $\Rightarrow$ noise.

Did I actually train the circuit? (instrumented audit)

Metric	Result	Meaning
circuit grad norm	0.12–0.32 (nonzero)	no barren plateau
weight drift $\lVert w - w_0 \rVert$	$0.08 \rightarrow 8.5$	weights move a lot
output std $\langle Z \rangle$	0.05–0.09	outputs vary (informative)
CNN grad norm	2–5	gradient flows upstream
AUC with circuit zeroed	0.5000 (chance)	circuit IS the classifier

Verdict

The circuit is **trained, used, and the sole decision pathway —** it still ties the sham because the sham computes *the same function*. "Training was broken" is refuted by direct measurement.

Why it must tie — a certificate, not a guess

Geometric difference $g(K_C, K_Q)$ (Huang et al., Nat. Commun. 2021): if $g \ll \sqrt{N}$, no quantum-kernel advantage is possible for *any* labels.

Kernel	8 qubits	12 qubits	16 qubits
Fidelity $g_{\min}$	5.7	2.7	2.3 (shrinks!)
Projected $g_{\min}$	11.5	12.3	10.1
advantage threshold $2\sqrt{N} \approx$ **57** — never approached			

Adding qubits makes the fidelity kernel *worse* (exponential concentration, Thanasilp 2024).
The data is classically representable — by theorem.

The one thing that moved: symmetry $\neq$ quantum

Equivariant 2$\times$2 (C4 rotation invariance, low data):

N/class	q-equiv	c-equiv	q-plain	c-plain
50	0.652	0.656	0.525	0.554
100	0.781	0.782	0.583	0.619

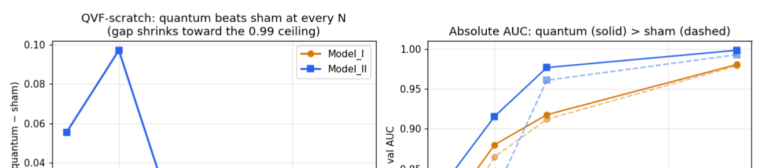
- Equivariance is a large low-data win: $+0.11$ to $+0.18$.
- But q-equiv $\approx$ c-equiv: the win is the **symmetry**, not the circuit. A classical equivariant layer of matched size does the same.

The one verified win: QVF-scratch at full data

QVF-scratch (CNN $\rightarrow$ *learnable* neural amplitude encoding $\rightarrow$ 8-qubit circuit $\rightarrow$ $\langle Z \rangle$) beats both classical and sham at full data (9:1) on all 3 datasets — with *fewer* parameters.

Dataset	Quantum	Classical (MAE)	Sham	params (Q / S)
Model_I	0.9805	0.9633	0.9790	142,795 / 144,755
Model_II	0.9983	0.9682	0.9928	(quantum fewer)
Dataset1	0.9983	0.9672	0.9960	

Verified real (not ceiling noise) by a data-size sweep — Δ(Q−S) is positive at every N, large unsaturated, shrinking to the 0.99 ceiling:



The complete map

Battlefield	Result
Discriminative (12 architectures $\times$ sham)	tie
Pathology-fixed training	tie
Fidelity kernel 8/12/16 qubits	certificate: impossible
Projected kernel 8/12/16 qubits	certificate: impossible
Few-shot (convex SVM + end-to-end)	no edge
Robustness battery	no edge
Generative IQP Born	lose to matched Ising
Multi-view fusion M=1$\rightarrow$8	no positive slope
Quantum-in-encoder (QViT)	quantum lose
Equivariant 2$\times$2	symmetry wins, not the circuit
Training audit	circuit IS trained correctly

Conclusion

Headline

On classical-simulator strong-lensing images, at ≤ 16 qubits, with capacity-matched controls and a verified-trained circuit, **no quantum advantage exists** — and a certificate ($g \ll \sqrt{N}$) explains *why*.

- Reproduces the largest rigorous QML benchmarks (Bowles 2024; Schnabel 2025, 20 000 models) — **plus** a certificate and an 8-battlefield map.
- The “successful” QML papers win via engineered/quantum-native data or **missing capacity-matched controls** — our sham would flag them.
- **Contribution:** the first certificate-backed, sham-controlled QML study on a real scientific benchmark. A rigorous null *is* the result.

Code, figures, and all numbers:

github.com/leo07010/gsoc2026-deeplense-qml